

2024



Hosting Medical Imaging Software Securely Using AWS with Siemens Healthineers

Learn how Siemens Healthineers in Spain collaborated with Overon, an AWS Partner, to connect multiple radiology centers of HT Médica across Spain using Amazon WorkSpaces.

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Up to 50%

reduction in customer spending on hardware and licensing

Hours instead of weeks

to set up new customers

Achieved

optimal latency

Improved

customer scale



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Overview

[Siemens Healthineers](#) needed a secure, accessible way to host its multimodality imaging software solution [on the cloud](#) to meet the needs of its longtime customer [HT Médica](#). HT Médica had built its own radiology information system, but it needed to connect geographically disparate radiology centers.



goals were to provide radiologists with better access to imaging, address a critical shortage in technicians, and offer the same radiology tests to customers no matter their location.

Siemens Healthineers chose to work with [Overon](#) (Part of GRUP MEDIAPRO), an [AWS Partner](#), to host syngo.via on Amazon Web Services (AWS) for HT Médica. The validation process took 1 year, and after centralizing its radiology information system, HT Médica saw an up to 50 percent reduction in licensing and hardware costs, achieved optimal imaging latency for its radiologists, and increased patient access to medical tests across Spain.



Opportunity | Using Amazon WorkSpaces to Host syngo.via on AWS for Siemens Healthineers

Siemens Healthineers, a medical technologies company headquartered in Erlangen, Germany, aims to pioneer better access to healthcare for patients worldwide by providing technology solutions for its customers' challenges. Siemens Healthineers had worked with HT Médica, a chain of 23 imaging centers across Spain, for 15 years and recognized the company as a leader in innovation in radiology technology. During one of their regular meetings, HT Médica expressed the desire to migrate to the cloud to alleviate pressure on its radiologists and reduce costs. Both companies recognized the importance of developing a secure solution that could maintain the privacy of patient information and comply with strict European regulations while expanding equity of access to high-quality radiology care across Spain.

To meet its customer's modernization goals, Siemens Healthineers chose to host the main solution it provided for HT Médica, syngo.via, on AWS. To achieve this, the company decided to work with Overon—a technology company within the Mediapro group that specializes in audiovisual content. "HT Médica needed a postprocessing and artificial intelligence layer to democratize access to it from every center," says Dr. Antonio Luna at HT Médica. "syngo.via on AWS has fulfilled these goals, keeping costs constrained."

Solution | Meeting Security Standards and Reducing Costs by up to 50% for HT Médica

On AWS, Siemens Healthineers deployed its solutions in a couple of hours rather than weeks. By centralizing and modernizing its radiology information system, HT Médica gained benefits for its business, employees, and patients. Local facilities benefited from a simple computer connectio —engineers no longer had to travel with expensive equipment to operate machines onsite, and facilities could purchase licenses without delay. “This solution is valuable to our customers because they can benefit from the technology without needing to create the infrastructure to facilitate it,” says David Alvarez, team lead for Spain at Siemens Healthineers. “With all the experience we gained and procedures we created hosting this solution on AWS, we can significantly accelerate adoption of the solution for new customers.”

Using the cloud-based version of Siemens Healthineers’ solution, HT Médica reduced its spending on hardware and licensing by 50 percent. The solution runs on [Amazon Elastic Compute Cloud](#) (Amazon EC2), which offers secure and resizable compute capacity for virtually any workload. The company could now scale Amazon EC2 GPUs to replace hardware costs. Given the breadth of Amazon EC2 instances to choose from, Siemens Healthineers worked closely with Overon to make use of the ones most applicable to its needs. The solution uses [Amazon EC2 C5 Instances](#), which are designed for compute-intensive workloads, and [Amazon EC2 G4 Instances](#), cost-effective GPU instances for machine learning inference and graphics intensive applications. Using Overon’s orchestration service, compute resources and workspaces are allocated according to the regular daily and weekly radiology schedule, and HT Médica can react to unplanned surges during off-hours or seasonal peaks. The system reacts to notifications sent from the patient scheduling tool, as well as to the needs of the medical staff accessing the system, avoiding overprovisioning and waste of compute resources.

To connect radiologists throughout Spain to the solution in the cloud, the solution uses [Amazon WorkSpaces](#), a service that offers all-inclusive, fully persistent virtual desktops for every worker. Using this service, radiologists can work remotely, reducing the demand to travel to distant sites. The high-resolution medical image display and low-latency desktops support radiologists’ ability to make diagnoses as if they were working on local machines. “Optimizing the network using Amazon WorkSpaces and the knowledge of Overon in adjusting certain parameters to decrease the latency as much as possible was crucial to the project,” says Alvarez.

To help meet security requirements, maintain data privacy, and meet other compliance needs common to medical groups, the solution uses additional AWS services, including [AWS Key Management Service](#) (AWS KMS), a service that provides the opportunity for users to create, manage, and control cryptographic keys to encrvnt or digitally sign data across applications and AWS services.



Delivery of the syngo.via solution in conjunction with Siemens Healthineers' artificial intelligence infrastructure solution previously took months but is now possible in days. The cloud-based imaging solution has increased the precision of radiologists' reporting, which has improved clinical outcomes for patients. "Using this solution, radiologists have access to state-of-the-art applications," says Saket Shukla, zone business lead for digital business for Southern Europe at Siemens Healthineers. "The solution scales up, which improves ease of access and leads to better diagnoses."

Outcome | Expanding Patient Access to Care with Scalable Imaging

Although the migration of syngo.via to AWS was the first collaboration between Siemens Healthineers and Overon, the two companies have since validated other Siemens Healthineers digital products and solutions using AWS cloud-based infrastructure. This solution is scalable and can be multitenant, meaning several hospitals can use a central infrastructure while keeping their data separate and secure. Siemens Healthineers plans to scale its solution for broader use to benefit even more patients.

"This marks a truly transformative time for cloud technology in the healthcare sector," says Dr. Razvan Ionasec, CTO of healthcare EMEA at AWS. "Over the past few years, we've witnessed a significant surge in the adoption of cloud-native imaging solutions on AWS, along with the migration of imaging workloads to the cloud for production. Today, healthcare providers around the world strive to combine medical images with omics data and health records to deliver insights for precision medicine. AWS provides purpose-built healthcare services to improve clinical efficiency and, ultimately, patient care."

About Siemens Healthineers

Siemens Healthineers is a medical technology company dedicated to expanding patient access to care worldwide through modern innovations, including virtual consultation services, multisite medical image sharing, and remote medical equipment management.

AWS Services Used

Amazon WorkSpaces

Amazon WorkSpaces Family solutions provide the right virtual workspace for varied worker types, from any location.

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